7.4 Exponential Change and Separable Differential Equations

Exponential functions increase or decrease very rapidly with changes in the independent variable. They describe growth or decay in many natural and human-made situations. The variety of models based on these functions partly accounts for their importance.

① Exponential Change

In modeling many real-world situations, a quantity y increases or decreases at a rate proportional to its size at a given time t. Examples of such quantities include the size of a population, the amount of a decaying radioactive material, and the temperature difference between a hot object and its surrounding medium. Such quantities are said to undergo exponential change.

If the amount present at time $t = 0$ is called y_0, then we can find y as a function of t by solving the following initial value problem:

$$\text{Differential equation: } \frac{\mathrm{d}y}{\mathrm{d}t} = ky$$

$$\text{Initial condition: } y(0) = y_0$$

If y is positive and increasing, then k is positive, and we say that the rate of growth is proportional to what has already been accumulated. If y is positive and decreasing, then k is negative, and we say that the rate of decay is proportional to the amount still left.

We see right away that the constant function $y = 0$ is a solution if $y_0 = 0$. To find the nonzero solutions, we divide Equation by y:

$$\frac{1}{y}\cdot\frac{\mathrm{d}y}{\mathrm{d}t} = k$$

$$\int \frac{1}{y}\,\mathrm{d}y = \int k\ \mathrm{d}t$$

$$\ln|y| = kt + C$$

$$y = A\mathrm{e}^{kt}.$$

- The solution of the initial value problem

$$\frac{dy}{dt} = ky,\ y(0) = y_0$$

is

$$y = y_0 e^{kt}.$$

Quantities changing in this way are said to undergo exponential growth if $k > 0$ and exponential decay if $k < 0$. The number k is called the rate constant of the change.

② General first-order Differential Equations

Exponential change is modeled by a differential equation of the form

$$\frac{dy}{dt} = ky,$$

where k is a nonzero constant. More generally, a first-order differential equation is an equation of the form

$$\frac{dy}{dx} = f(x, y),$$

where f is a function of *both* the independent and dependent variables.

- A **solution** of the equation is a differentiable function $y = y(x)$ defined on an interval of x-values such that

$$\frac{dy(x)}{dx} = f(x, y(x)).$$

- The **general solution** is a solution $y(x)$ that contains all possible solutions and it always contains an arbitrary constant.

③ Separable Differential Equation

The equation $\frac{dy}{dx} = f(x, y)$ is **separable** if f can be expressed as a product of a function of x and a function of y.

Example 1 Solve $y\frac{\mathrm{d}y}{\mathrm{d}x} = (1 + y^2)\mathrm{e}^x$.

Example 2 Solve $\frac{\mathrm{d}y}{\mathrm{d}x} = (1 + y)\mathrm{e}^x, y > -1$.

Example 3 Solve $y(x + 1)\frac{\mathrm{d}y}{\mathrm{d}x} = x(1 + y^2)$.

Example 4 Solve $\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2\mathrm{e}^{-y}$.

Example 5 Solve $\sqrt{x}\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{e}^{y+\sqrt{x}},\ x > 0$.